PV and Battery Trading in MGP and MI Auctions
Current Mathematical Model and Execution Workflow

`bess_pv_trading` repository

Implementation audited on 3 November 2026

Abstract

This report documents the implemented decision and execution model for a photovoltaic plant with a shared point of interconnection and a hypothetical AC-coupled battery. It specifies the Italian day-ahead and three intraday auction timeline used by the code, the residual mixed-integer optimization, immutable auction commitments, battery degradation, interval-level feedback controller, optional real-time economic MPC, and ex-post imbalance settlement. It distinguishes the general scenario-tree interface from the deterministic actual-price oracle used in the historical studies. The report follows current source code and shared YAML defaults; it is not a market strategy validation or an investment valuation. PV forecasting research is summarized only as an input dependency, with the selected farm bundles identified for reuse by the continuous optimization runner.

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1 Scope and implementation map

One farm has PV generation, an AC-coupled BESS, and a common AC grid connection. There is no local electricity demand. PV can be exported, stored, or curtailed; the battery can charge from PV or grid imports and discharge toward the grid. The market sequence is MGP (called DA in code), MI-A1, MI-A2, and MI-A3 (IDA1, IDA2, IDA3). Every submitted signed quantity is assumed to execute in full at the auction clearing price. Bid prices, acceptance uncertainty, market impact, and continuous intraday trading are outside this formulation.

The implementation keeps four distinct objects:

1. An optimization plan, which includes the current actionable auction order, future auction recourse, and a battery/PV schedule over the remaining horizon.
2. An optional real-time MPC plan over fixed submitted commitments, with a separately recorded first executable physical action.
3. An immutable commitment ledger, updated at submission and publication.
4. A physical trajectory produced by a feedback/protection policy using realized interval-average PV. Its energy, wear, and deviations are settled separately from the optimization objective.

Source	Responsibility
`src/domain.py`	Validated grids, assets, scenarios, information nodes, states and solver contracts.
`src/optimization/model.py`	Physical, trading, product-resolution, optional imbalance constraints and objective.
`src/optimization/mpc.py`	Common-control real-time MILP and budget-constrained imbalance-price oracle.
`src/optimization/physical.py`	Shared AC battery equations, exclusive controls and degradation.
`src/optimization/pv_envelope.py`	Conditional exact-delivery PV-envelope reduction and reconstruction of original member results.
`src/optimization/degradation.py`	Throughput coefficient and segmented cycle-depth model.
`src/optimization/validation.py`	Numerical validation independent of the solver constraint objects.
`src/transaction_fees.py`	Pure-Python fee integration, dated tariffs and operator monthly state.
`src/optimization/transaction_fees.py`	Flat-band LP costs and exact progressive ABS/PWL constraints.
`src/simulation/synthetic.py`	Shared event replay and auction submission/publication accounting.
`src/simulation/replay.py`	Residual scenario conditioning and bounded physical execution.
`src/simulation/gme_oracle.py`	Manual-data historical price oracle, rolling PV and full-information asset comparisons.
`src/simulation/continuous.py`	Consecutive delivery-day books, shadow boundary projections and uninterrupted execution.
`src/simulation/continuous_inputs.py`	Offline input preflight, deterministic forecast snapshots and optional frozen weather ensembles.
`src/simulation/realtime_control.py`	Interval-start MPC actions and deterministic open-loop shadow controls.

Source	Responsibility
`src/simulation/realtime_mpc_study.py`	Frozen-ledger and continuous comparisons using the same event runner.
`src/reporting/realtime_mpc.py`	Paired controller metrics, plans, signed cash accounting and plots.
`src/reporting/continuous_outputs.py`	Continuous trajectory accounting, paired comparisons and native solve timing.
`src/simulation/settlement.py`	Exact-grid actual settlement and expected/realized KPI reconciliation.
`src/forecasting/models/pv.py`	Weather-to-AC PV conversion.
`src/markets.py`	Configurable local auction gates, publication events and DST-aware delivery grids.

2 Time, units, signs and information

2.1 Delivery intervals and auction products

A local delivery day is partitioned into physical intervals $\mathcal{T} = \{0, \dots, T-1\}$, with starts τ_t, ends τ_{t+1} and durations

$$\Delta_t = (\tau_{t+1} - \tau_t)/\text{one hour}. \tag{1}$$

The historical study uses quarter-hours, so $\Delta_t = 0.25$. A local day is constructed between successive midnights in `Europe/Rome`, then stored in UTC: T is 92, 96, or 100 for spring-DST, ordinary, or autumn-DST days. Repeated local hours are distinguished by their actual UTC offset/fold. No equation assumes that every day has exactly 24 elapsed hours.

A physical interval is not necessarily an independent market product. Before 1 October 2025, DA products are hourly; the MI auctions in the current 2025 history are already quarter-hourly. Physical battery controls are quarter-hourly at all four optimization stages. For the four equal-duration quarters of one native DA hour h, the implementation enforces

$$q_{\mathrm{DA},n,t} = q_{\mathrm{DA},n,t'} \quad (t, t' \in h). \tag{2}$$

These are equal MWh shares of one hourly order, whose total is $\sum_{t \in h} q_{\mathrm{DA},n,t}$. The hourly clearing price is repeated as an objective coefficient on those shares. The quarters do not become separate DA trades or separate observed prices. There is no corresponding date-based hourly equality on battery charging or discharging. Native product resolution and optional physical-control grouping are separate configuration fields.

The configured auction calendar uses local `Europe/Rome` time. The shared calendar validity interval is 1 October 2025 through 31 August 2026; historical study configurations may explicitly override it. The decision cutoff is five minutes before each gate in the supplied studies.

Auction	Gate	Decision cutoff	Results clock	Delivery scope
MGP/DA	D−1 12:00	D−1 11:55	D−1 13:30*	Whole day D
MI-A1/IDA1	D−1 15:00	D−1 14:55	by D−1 15:30	Whole day D
MI-A2/IDA2	D−1 22:00	D−1 21:55	by D−1 22:30	Whole day D
MI-A3/IDA3	D 10:00	D 09:55	by D 10:30	D 12:00–24:00

*The configured 13:30 DA publication is a conservative study assumption; GME's technical rule says final MGP results are not published earlier than 13:05. Recorded publication timestamps override the configured clock.

MI-A1/A2/A3 are the three auction sessions; they exclude the separate continuous MI-XBID market. Every submitted quantity is assumed accepted in full and settled at the auction clearing price. The model does not optimize bid prices, acceptance probabilities, market impact or partial fills. DA products were hourly in the studied data through September 2025. GME made 15-minute MGP products available for delivery from 1 October 2025, and also announced 30-minute products. MI auction products in the studied 2025 data were already quarter-hourly. Product-duration groups, rather than a hard-coded number of variables per day, define trade variables and native-product equalities. The delivery grid is always constructed between local midnights and stored in UTC, so DST days can contain 92, 96 or 100 quarter-hours.

An auction a has an eligibility set $\mathcal{T}_a$. A documented/configured cancellation removes that auction from the active set. Missing data alone does not establish a cancellation: unexplained missing or incomplete curves fail validation. Absent auctions create no trade variables and no submission solve.

At DA closure the integrated model may optimize planned DA, IDA1, IDA2 and IDA3 quantities together. At IDA1, DA is fixed and the remaining three markets may be planned; at IDA2, DA and IDA1 are fixed; at IDA3, the first three are fixed and only IDA3 quantities are free. This integrated planning uses actual future auction curves in the price-oracle experiments, although those prices would not be known at the earlier real-world gates. The sequential baseline removes future-auction trade variables and their expected revenue from each solve. Both modes replan physical operation over the remaining eligible horizon and preserve earlier commitments.

2.2 Notation and electrical basis

Symbol	Unit	Meaning
E	MWh	Nominal internal stored-energy capacity.
$E^{\min}, E^{\max}$	MWh	Operational energy limits derived from SOC percentages.
$\overline{P}$	MW AC	One symmetric battery charging/discharging rating.
$P^{\text{imp}}, P^{\text{exp}}$	MW AC	Directional shared POI ratings.
η_c, η_d	fraction	Complete AC-bus-to-storage and storage-to-AC efficiencies.
σ	fraction/hour	Configured hourly self-discharge fraction.
$V_{s,t}$	MW AC	Forecast available PV in scenario s.
$\widetilde{V}_t$	MW AC	Effective realized PV used for execution.
$c_{n,t}, d_{n,t}$	MW AC	Battery charge/discharge commands at information node n.
$u_{n,t}$	MW AC	PV curtailment.
$i_{s,t}$	MW AC	Uncurtailed PV injection into the common AC bus.
$g_{s,t}$	MW AC	Net grid exchange; positive is export, negative is import.
$e_{s,t}$	MWh	Internal battery energy at an interval boundary.
$q_{a,n,t}$	MWh	Signed auction adjustment; positive sale, negative purchase.
$Q_{s,t}$	MWh	Final cumulative contracted delivery position.
$\pi_{a,s,t}$	EUR/MWh	Auction-price coefficient for interval t.
w_s	fraction	Scenario probability, summing to one.

Power variables are AC at the common bus/POI. Battery energy is internal stored energy. Farm nameplate PV capacity is AC. For pvlib, the DC reference rating is derived as

$P_{\mathrm{DC,STC}} = P_{\mathrm{AC,rated}}/\eta_{\mathrm{ref}}$, where the reference factor is a separate, fixed configured assumption (currently 0.96). The physical PVLib path applies the configured DC-to-AC and transformer conversions and caps output at the farm AC rating. Saved neural bundles instead produce an AC prediction subject to their own output bounds and the configured night policy; they do not receive an additional PVLib loss conversion. Measured historical `power` is already AC kW and is converted once to MW; no PV or transformer efficiency is applied to it. BESS efficiency is not applied to measured PV or direct PV exports.

2.3 Forecast vintages and observability

A weather forecast cycle v has source time t_v and availability time A_v. Without explicit source publication times, the configured convention is $A_v = t_v + 6$ hours. At decision cutoff k, the rolling policy chooses the latest eligible cycle with $A_v \le k$ and required remaining-horizon coverage. Weather cycles about six hours apart are successive forecast vintages, not ensemble members. The deterministic weather store supplies one forecast member; separate ensemble archives can additionally supply multiple member trajectories.

Measured interval-average PV becomes available after interval end plus configured telemetry delay. A forecast for future intervals is not replaced by future realized power in the ordinary auction replay. If a decision occurs inside an unfinished quarter-hour, the next boundary state is projected through that interval with forecast PV and the same execution policy. Its final realized average cannot yet be observed. The hindsight counterfactuals deliberately have future PV and price information and are labeled as such.

Policy-treated observations carry evidence times. Plateau evidence can require later samples; two-sided interpolation requires its right endpoint and applicable mask evidence. A finalized corrected time series therefore differs from the as-of view available to a past decision. This remains relevant even when a study uses actual-price foresight.

3 PV availability supplied to the optimizer

The optimizer receives nonnegative AC power trajectories $V_{s,t}$ for the represented scenarios. The deterministic farm weather store contains vintages keyed by farm, UTC cycle and hourly lead; successive cycles are not ensemble members. Separate weather-ensemble archives include an explicit member axis. Forecast cycles are selected using publication availability, not just initialization time. The configured availability lag is six hours. Hourly radiation values are aligned to the quarter-hour delivery grid according to their interval-average semantics. PV output is capped at the declared farm AC rating. Historical measured power is already AC and receives no additional conversion loss.

The reference physics path uses configured GHI and horizontal beam irradiance, pvlib solar geometry, fixed-array metadata or the configured single-axis tracker model, plane-of-array irradiance, Faiman module temperature and PVWatts DC output. In symbolic form,

$$G_{\mathrm{DHI}} = G_{\mathrm{GHI}} - G_{\mathrm{BHI}}, \qquad G_{\mathrm{DNI}} \simeq G_{\mathrm{BHI}}/\cos\theta_z, \tag{3}$$

$$G_{\mathrm{POA}} = G_{\mathrm{beam}} + G_{\mathrm{sky}} + G_{\mathrm{ground}}, \qquad T_{\mathrm{mod}} = T_{\mathrm{air}} + \frac{G_{\mathrm{POA}}}{U_0 + U_1 v}, \tag{4}$$

$$P_{\mathrm{DC}} = \max\left(0, \frac{P_{\mathrm{AC,rated}}}{\eta_{\mathrm{ref}}} \frac{G_{\mathrm{POA}}}{1000}[1 + \gamma(T_{\mathrm{mod}} - T_{\mathrm{ref}})]\right). \tag{5}$$

The shared Faiman coefficients are $U_0 = 25$ and $U_1 = 6.84$, and the configured PVWatts coefficient is $\gamma = -0.004/^{\circ}\mathrm{C}$. Historical fit artifacts retain the settings under which they were produced. The DC reference ratio $\eta_{\mathrm{ref}} = 0.96$ is distinct from the configured DC-to-AC scale. The final AC

forecast uses the selected output coefficient, transformer setting and AC cap. Farm coordinates are a documented proxy, so the physical geometry is an assumption. The optimization does not use satellite GHI as an operational weather forecast; it is retained for retrospective diagnosis.

The modeling-quality mask for model training and evaluation has one production authority: `configs/data_quality/pv_excluded_episodes_2025_to_2026_08.csv`. Complete feature/target rows identified there are removed from model splits when the configured exclusion policy is enabled; sequence models do not bridge removed quarters. Those episode exclusions are not automatically applied to the measured PV used by auction execution. The shared `policy.realized_pv_night` instead sets power to zero outside configured month-specific local production windows; the window is applied to each interval's local start, and missing nighttime measurements are also set to zero. The raw store is not modified. `policy.forecast_pv_night` applies the same monthly prior to deterministic and ensemble PV inference, regardless of model. Missing realized PV is handled separately using the configured frozen-forecast fallback and missing-interval cap. Provenance records clipping and fallback.

3.1 Selected farm bundles for continuous optimization

The per-farm pilot registry points to self-contained bundles. These are loaded only when `pv.forecast_replacement.enabled:true`; the default remains the pvlib path. For the deterministic forecast, a missing or not-yet- available configured artifact uses the declared PVLib fallback. The ensemble preflight instead requires an existing supported weather-bundle artifact when one is selected; an explicitly null farm entry selects PVLib. All models were trained on 2025 data; pilot choice used a previously inspected 2026 report period, so it is retrospective rather than an as-known-at-the-time benchmark.

Farm	Selected model	Timing channels	Bundle path
1	Direct TCN 02	none	`models/pv/farm_1/pilot_2025_train/direct_tcn/`
345	Direct TCN 12	none	`models/pv/farm_345/pilot_2025_train/direct_tcn/`
901	Direct TCN timing 05	lead and cycle hour	`models/pv/farm_901/pilot_2025_train/direct_tcn/`
910	Direct TCN paired timing	lead and cycle hour	`models/pv/farm_910/pilot_2025_train/direct_tcn/`

The table describes the shared registry, not an unconditional model selection. The current four-farm January and August deterministic/ensemble study overrides select the registry bundles for farms 1, 901 and 910. For farm 345 they explicitly select PVLib with the shared unfitted output/temperature defaults and a Genoa location proxy (44.4056 N, 8.9463 E); the proxy is not a verified site location. Other study configurations, including the earlier farm-1 PVLib scenario studies, may use the baseline instead. The shared replacement map points to each bundle's `manifest.json`; the continuous runner verifies its farm identity and checksums and preserves the trained feature preprocessing. The default model availability timestamp is 27 October 2026 UTC. The January forecast-pilot configurations explicitly allow retrospective use. `experiment-continuous` consumes these bundles; the historical `experiment-gme-oracle` path currently uses its configured pvlib forecast. Model details and standalone inference instructions are in `models/pv/README.md`.

3.2 Optional frozen weather ensemble

When `continuous_optimization.pv_ensemble.enabled` is true, preflight reads the preceding delivery day's 00 UTC ensemble cycle, assumes availability after the configured lag, and requires it

to be available at the first cutoff. The cycle is frozen for that delivery day. Each weather member is converted separately to AC PV using the configured PVLib path or a supported weather-bundle model; no ensemble calibration, reweighting or post-processing is implicit. By default, the cutoff-updated deterministic forecast is prepended as another member. If the archive has 50 members, the scenario count is 51 and each weight is 1/51; the optional weather-only sensitivity instead has 50 equal weights. The deterministic member can change at each gate, and at each selected MPC cutoff, while the other 50 paths remain fixed over the day. The weather-bundle inference records its model artifact and retrospective availability; the six-hour historical ensemble availability is an assumption, not a measured publication timestamp. Ensemble bundle inference labels the frozen paths with the DA auction context; the deterministic member is evaluated for the current auction at each gate.

In this PV-scenario mode, auction orders and planned battery commands are shared across members; scenario-specific curtailment provides interval PV recourse, with equal observed PV tied to the same action. Planning uses the same actual auction price curves across members in the historical oracle studies. The average objective includes any scenario-dependent imbalance terms when they are enabled; in exact-delivery mode shared schedules and prices make the member operating objectives equal. In a scenario formulation, the execution snapshot takes curtailment from the first scenario; the existing grid-tracking controller adapts it to realized PV. Full aligned member values, vintages and input fingerprints are saved separately.

The optional `ensemble_mean` policy uses the pointwise arithmetic mean of the weather members only, including for locked-prefix projections. The optional `ensemble_stochastic` policy preserves the full scenario formulation and replaces exact planned delivery by scenario deviations $\delta_{s,t}$. It maximizes expected operating profit less

$$c \sum_s p_s \sum_t |\delta_{s,t}| + \rho \left(z + \frac{1}{1-\alpha} \sum_s p_s u_s \right), \qquad u_s \geq \sum_t |\delta_{s,t}| - z, \quad u_s, z \geq 0. \tag{6}$$

Here c and ρ are configured EUR/MWh weights, and α is CVaR confidence. The CVaR weight defaults to zero. The stochastic planning imbalance price is zero, rather than a future realized settlement price. The objective uses the configured idealized scenario-dependent curtailment recourse; this is not an optimized real-time controller or an intraday recourse tree. The first scenario is the deterministic member when included; in the weather-only sensitivity it is the first weather member. The unchanged feedback controller executes its planned battery commands and adjusts curtailment against actual PV.

4 Optimization at an auction decision

Each residual solve uses only the remaining suffix of the delivery grid. In the equations below that suffix is reindexed from zero, with T denoting its final boundary; its original UTC product identifiers are retained for commitments.

4.1 Scenario tree and nonanticipativity

At decision k, let $\mathcal{S}_k$ be represented leaf scenarios and let $\mathcal{A}_k$ be the open active auctions. An information tree assigns an auction bid owner $n_a(s)$ to each auction/leaf and a control owner $n_t(s)$ to each interval/leaf. Curtailment may have a separate owner $m_t(s)$ under the selected execution-information convention.

Decisions are node-owned variables, not independent leaf decisions followed by a penalty for disagreement. For example,

$$c_{s,t} = c_{n_t(s),t}, \quad d_{s,t} = d_{n_t(s),t}, \quad q_{a,s,t} = q_{a,n_a(s),t}, \quad u_{s,t} = u_{m_t(s),t}. \tag{7}$$

Thus two scenarios with the same observable history share actionable decisions. A latent PV-member label is not available information by itself. Constraints apply to every represented leaf, including zero-probability leaves if supplied; probabilities only weight the objective. In the current one-scenario historical price oracle, the tree degenerates to deterministic planning with known future prices and a forecast PV trajectory.

4.2 Exact-delivery PV envelope reduction

For a multi-member input with exact interval delivery, common auction orders and battery commands, identical member prices and compatible instantaneous PV-curtailment owners, the optimizer uses one pointwise-minimum PV trajectory. The common position and battery dispatch imply identical PV injection i_t in every member, so $0 \le i_t \le \min_s V_{s,t}$. Conversely, an envelope solution can be lifted by setting $u_{s,t} = V_{s,t} - i_t$ for each member. Physical costs, trades and objective remain unchanged; zero-probability members still constrain the minimum. The solver reconstructs the original scenario-shaped solution and independently validates it. The full model is retained when these conditions do not hold, including when planned imbalances are enabled. Diagnostics record the original and modeled member counts. This reduction changes neither the weather inputs nor real-time feedback, and equivalence at a fixed planning state does not guarantee identical trajectories when alternative optima are selected. The farm-1 pilot compared the full model and envelope at 37 matched auction states across 2–11 August 2026, then ran each as an independent continuous trajectory under the curtailment-recovery controller. The recorded objectives and realized trajectories agreed to numerical precision on that sample; this is empirical support, not an assertion that independent solvers always choose the same optimum. The study design and limitations are in `docs/pv_lower_envelope_replay.md`.

4.3 Physical variables and balances

For every scenario and remaining interval:

$$i_{s,t} + u_{s,t} = V_{s,t}, \qquad i_{s,t} \ge 0, \qquad 0 \le u_{s,t} \le V_{s,t}, \tag{8}$$

$$g_{s,t} = i_{s,t} + d_{s,t} - c_{s,t}, \tag{9}$$

$$-P^{\text{imp}} \le g_{s,t} \le P^{\text{exp}}. \tag{10}$$

Curtailment appears only in the PV balance. It cannot erase bought grid energy or battery charging energy. A negative grid exchange must be absorbed by net battery charging since there is no local demand. The model does not explicitly assign charging power to its PV/grid source; that partition is an accounting convention used in post-processing.

With $z_{n,t} \in \{0, 1\}$ denoting charging mode,

$$0 \le c_{n,t} \le \overline{P} z_{n,t}, \qquad 0 \le d_{n,t} \le \overline{P}(1 - z_{n,t}). \tag{11}$$

This excludes simultaneous charge and discharge, uses one symmetric AC rating, and introduces no separate battery-inverter power limit.

SOC percentage parameters $s_{\min}, s_{\max}$ define

$$E^{\min} = E\frac{s_{\min}}{100}, \qquad E^{\max} = E\frac{s_{\max}}{100}, \qquad E^{\min} \le e_{s,t} \le E^{\max}. \tag{12}$$

For $\rho_t = (1 - \sigma)^{\Delta_t}$, the transition is

$$e_{s,t+1} = \rho_t e_{s,t} + \eta_c c_{s,t} \Delta_t - \frac{d_{s,t}\Delta_t}{\eta_d}. \tag{13}$$

One AC MWh charged stores η_c MWh. Delivering one AC MWh consumes $1/\eta_d$ internal MWh. In combined mode the two configured efficiencies each cover their entire directional path. In converter-and-cell mode, $\eta_c = \eta_{c,\text{conv}}\eta_{c,\text{cell}}$ and $\eta_d = \eta_{d,\text{conv}}\eta_{d,\text{cell}}$. The alternatives cannot be mixed. The round-trip factor is $\eta_c\eta_d$. Self-discharge acts on stored energy, not on a power variable, and is separate from cycle degradation cost.

The initial boundary is fixed to measured/carried/projected state e^k:

$$e_{s,0} = e^k. \tag{14}$$

If the terminal target is configured, also impose

$$e_{s,T} = e^{\text{target}}. \tag{15}$$

In the GME adapter, initial/terminal percentages are converted using nominal capacity. A `null` terminal SOC omits (15). Prepared and synthetic boundary-state contracts use explicitly named MWh values. Neither version adds a terminal energy salvage value to the objective. An independent optional lower target, configured by `state.terminal_soc_min_percent`, imposes

$$e_{s,T} \geq E\, s_{\text{floor}}/100. \tag{16}$$

Its shared default is `null`. When enabled it applies to every planning horizon end, including intermediate delivery days in continuous operation. A non-null final equality and this floor both apply on the final day. The floor constrains planned energy; forecast-error feedback can produce a realized closing SOC below it while still respecting the hard SOC box.

4.4 Auction quantities and physical delivery

Let Q_t^{fix} contain all previously submitted commitments, priced or still pending. The new final position on leaf s is

$$Q_{s,t} = Q_t^{\text{fix}} + \sum_{a \in \mathcal{A}_k :\, t \in \mathcal{T}_a} q_{a,n_a(s),t}. \tag{17}$$

An ineligible auction/interval has no trade variable. The model allows signed sales/purchases; they are adjustments to prior positions, not gross generation. For every eligible trade,

$$-P^{\text{imp}}\Delta_t \leq q_{a,n,t} \leq P^{\text{exp}}\Delta_t. \tag{18}$$

For every open auction a in chronological order, define the cumulative planned position after that auction as

$$C_{s,t}^a = Q_t^{\text{fix}} + \sum_{b \in \mathcal{A}_k :\, b \leq a,\, t \in \mathcal{T}_b} q_{b,n_b(s),t}. \tag{19}$$

It satisfies

$$-P^{\text{imp}}\Delta_t \leq C_{s,t}^a \leq P^{\text{exp}}\Delta_t, \qquad -P^{\text{imp}}\Delta_t \leq Q_{s,t} \leq P^{\text{exp}}\Delta_t. \tag{20}$$

There is no independent commercial volume cap. The directional POI ratings apply to individual auction adjustments, cumulative positions after each auction, and physical delivery. An offsetting later trade is allowed, subject to its own POI-derived limit and the resulting position limit. Thus the model permits bounded cross-auction spread trading even without a battery; its revenues can be present in both asset counterfactuals.

With the shared default `trading.optimize_imbalances: false`, planned delivery is exact in every interval:

$$g_{s,t}\Delta_t = Q_{s,t}. \tag{21}$$

No positive/negative imbalance variables or imbalance direction binary are created. Forecast deviations are not an additional dispatch resource. An optional configured product-group delivery equality can additionally impose $\sum_{t \in h}(g_{s,t}\Delta_t - Q_{s,t}) = 0$. It is redundant with (21) and does not replace it in the disabled-imbalance mode.

4.5 Current objective and sunk cash flows

For throughput degradation with coefficient c_{thr} and an optional import fee f, the current residual objective is

$$\max J_k = \sum_{s \in \mathcal{S}_k} w_s \left[\sum_{a \in \mathcal{A}_k} \sum_{t \in \mathcal{T}_a} \pi_{a,s,t} q_{a,n_a(s),t} - c_{\text{thr}} \sum_t (c_{s,t} + d_{s,t}) \Delta_t - f \sum_t [-g_{s,t}]_+ \Delta_t - C_s^{\text{trading}} \right]. \tag{22}$$

The implementation uses an exact import-power linearization with a direction binary only when f is nonzero. The shared default sets this additional import fee to zero; market transaction fees are represented separately below.

4.6 Market transaction fees and operator monthly volume

The `transaction_fees.mode` is `none`, `flat`, or `gme_real` (the shared YAML default). Legacy programmatic inputs without a fee section remain fee-free. Let $m(a)$ be the calendar month of auction submission in market local time, V_m known already executed operator volume, and $G_{m,s} = \sum_{a:m(a)=m,t} |q_{a,n_a(s),t}|$ new gross MWh. Purchases and sales count separately even when final delivery is zero. Quarter-hour energy shares of an hourly product are summed once, without another duration factor. For a cumulative progressive tariff F_m,

$$C_s^{\text{trading}} = \sum_m \left[F_m(V_m + G_{m,s}) - F_m(V_m) \right]. \tag{23}$$

In flat mode this is the configured EUR/MWh coefficient times gross MWh. The published GME MPE marginal rates are 0 on the first 20,000 MWh/month, 0.04 up to 1,000,000, 0.03 up to 10,000,000, and 0.02 thereafter. These bands are progressive and not a retroactive rate on the monthly total.

Fee-free horizons introduce no auxiliary variables. Positive single-band costs use linear absolute-value epigraphs. Across reachable breakpoints, exact Gurobi ABS general constraints and a PWL equality represent both gross volume and the nonconvex tariff; a convex epigraph alone is not used for regressive bands. Absolute-value variables retain auction-node ownership and native-product sharing. Costs are evaluated for each leaf and weighted by w_s.

The fee state includes pending guaranteed transactions immediately at submission. Publication adds no second fee. Executed gross volume remains in V_m during residual solves, while incurred EUR are sunk. The state carries an availability clock and declared provenance; nonzero future executed monthly volume is rejected. Months use transaction dates, so pre-DA/IDA1/IDA2 and IDA3 for one delivery day may belong to different months. A new month starts at zero. The saved final fee state can initialize a later chronological study. Operator scope must explicitly distinguish a standalone plant from a portfolio with exogenous starting volume. Unknown future external activity is not invented: costs are exact conditional on the supplied initial state, not a complete operator-level monthly forecast.

Tariffs have explicit nonoverlapping complete-month validity windows, version, source, thresholds, marginal rates and an applicability status. The current GME tariff and the 2006 annual report show the same structure, the latter effective in 2007. A dated confirmation of 2025 applicability was not established; the shared December 2024–December 2026 tariff marks this explicitly as `assumed_unchanged`. Unsupported dates fail. The shared initial state assumes a standalone simulated operator at local midnight on 1 December 2024, with zero December volume, zero earlier fees and no external portfolio activity. Subsequent months begin at zero and continuous execution carries actual submitted volume. Known operator-level volume should replace this assumption

when the plant belongs to a larger portfolio. Enabling this tariff can still produce zero variable fees below the monthly exemption. Example YAMLs are `configs/full_experiments/synthetic_fees_none.yaml`, `configs/full_experiments/synthetic_fees_flat.yaml`, and `configs/full_experiments/synthetic_fees_gme.yaml`. Membership, taxes, ACER record charges and separate aggregator commissions remain excluded; physical grid fees and degradation remain separate cost components.

Past auction cash flow and already incurred degradation are sunk at residual reoptimization. Priced commitments remain in the external ledger, and pending unpriced submitted quantities enter Q^{fix} without adding an optimizable cash-flow term. Their future clearing value is constant with respect to all remaining decisions and is excluded from J_k. This avoids double counting. The initial pre-DA solve has no prior commitments; its objective spans the modeled whole-day opportunity set. Residual objective values therefore should not be summed across auction solves to compute daily profit.

There is no explicit curtailment penalty, risk measure, capital cost, fixed O&M, tax, brokerage fee, ramp constraint, minimum dwell time, or calendar-aging charge in (22). Curtailment can affect revenue through delivery choices, but is not independently priced. The horizon can exploit inventory when its terminal target is free; this matters when comparing asset value.

4.7 What is free at each of the four decisions

Decision	New free auction quantities	Previously submitted fixed quantities
DA	DA, IDA1, IDA2, IDA3 plans	None
IDA1	IDA1, IDA2, IDA3 plans	DA
IDA2	IDA2, IDA3 plans	DA and IDA1
IDA3	IDA3 eligible products	DA, IDA1 and IDA2

At each decision, remaining-horizon charge/discharge modes, curtailment, grid powers and energy states are free subject to their balances, boundary state, nonanticipativity, product rules and terminal condition. Executed past intervals are absent from the residual grid. Future-auction quantities are plans until their own submission. Only the current auction order is committed from that solve. Before noon, IDA3 cannot change orders for the morning; physical control commands from the previous plan also remain in force. All IDA3 decision variables cover only the 48 quarter-hours from 12:00 to midnight: charge, discharge, mode, curtailment, grid power and trades, with 49 SOC boundaries. Gate closure determines the information cutoff, not the delivery horizon.

More generally, let c_k be the submission cutoff and j_k the first grid boundary no earlier than both c_k and the current auction's first eligible delivery. The solver starts at j_k. From the latest observed physical state, the replay projects the locked prefix using the previously requested commands $\bar{u}_i$, fixed contracted grid targets Q_i/Δ_i, and the latest cutoff-available PV forecast $\widehat{P}_i^{(c_k)}$ through the same physical controller Φ used for delivery:

$$\widehat{x}_{i+1|c_k} = \Phi\left(\widehat{x}_{i|c_k}, \bar{u}_i, Q_i/\Delta_i, \widehat{P}_i^{(c_k)}\right), \qquad i < j_k. \tag{24}$$

Here x includes energy and degradation state. The resulting $\widehat{E}_{j_k|c_k}$ is the fixed initial energy of the residual MILP. Forecast publication eligibility is still checked at c_k; projecting to noon does not permit a forecast released after the 10:00 gate. No future measured PV is used in this projection. Actual delivery continues with the retained commands and realized PV, so its noon state may differ from the projected state. Grid feedback and protection enforce the actual SOC and power bounds. If IDA3 is the first participating auction, zero-position feedback continues until noon.

5 Battery degradation models

5.1 Current AC throughput approximation

For one trajectory,

$$H = \sum_t (c_t + d_t)\Delta_t, \qquad D_{\text{thr}} = c_{\text{thr}} H. \tag{25}$$

Both directional AC throughputs are charged once. The YAML can directly supply c_{thr} or derive it from replacement economics and reference cycle life. Let p_{USD} be pack USD/kWh, x be USD per EUR, p_{extra} additional EUR/kWh replacement cost, s_R salvage fraction, N reference cycle life and d_{ref} reference internal depth of discharge. Then

$$R = 1000E\left(\frac{p_{\text{USD}}}{x} + p_{\text{extra}}\right)(1 - s_R), \tag{26}$$

$$H_{\text{life}} = N d_{\text{ref}} E\left(\frac{1}{\eta_c} + \eta_d\right), \tag{27}$$

$$c_{\text{thr}} = \frac{R}{H_{\text{life}}}. \tag{28}$$

One reference internal cycle charges $d_{\text{ref}}E/\eta_c$ AC MWh and discharges $d_{\text{ref}}E\eta_d$ AC MWh, explaining the normalization. There is no additional factor of two or half applied to (28). The configured end-of-life capacity fraction is descriptive metadata, not an extra multiplier or a dynamic capacity update.

The shared YAML uses a historical generic pack-price proxy, a dated FX conversion, and an illustrative reference cycle life. Those are study inputs with source metadata, not an estimate of a specific installed BESS cost. The resolved R and c_{thr} are saved. The cost is financial wear; it does not reduce E, $\overline{P}$, or efficiencies as the simulation proceeds.

5.2 Optional segmented depth-dependent aging

The optional `xu_segmented` mode follows the configured segmented cycle-depth construction described in the repository's Xu notes and the author paper [1]. It is distinct from the current throughput study. For a configurable convex stress curve $\Phi(d) = \alpha d^\beta$, $\beta \geq 1$, partition nominal internal capacity into J equal segments of size E/J. The marginal AC discharge cost of segment $j = 1, \ldots, J$ is

$$m_j = \frac{RJ}{\eta_d E}\left[\Phi(j/J) - \Phi((j-1)/J)\right]. \tag{29}$$

It is nondecreasing with depth. The degradation charge is

$$D_{\text{Xu}} = \sum_t \sum_{j=1}^{J} m_j d_{j,t} \Delta_t. \tag{30}$$

Only discharging is charged under this convention. A discharging half-cycle incurs the convention's depth cost, while a charging half-cycle does not; do not add an extra factor of one half. Nonzero throughput cost is rejected in this mode.

Segment flows and energies obey

$$\sum_j c_{j,t} = c_t, \qquad \sum_j d_{j,t} = d_t, \qquad \sum_j h_{j,t} = e_t, \tag{31}$$

$$h_{j,t+1} = h_{j,t} + \Delta_t(\eta_c c_{j,t} - d_{j,t}/\eta_d), \qquad 0 \leq h_{j,t} \leq E/J. \tag{32}$$

Segment energies represent cycle-depth memory, not fixed absolute SOC buckets. Charging and discharging both use shallow available segments first. For each $j < J$, the implementation adds priorities $v^c_{j,t}, v^d_{j,t} \in \{0, 1\}$:

$$\sum_{\ell>j} c_{\ell,t} \le \overline{P} v^c_{j,t}, \qquad h_{j,t+1} \ge (E/J) v^c_{j,t}, \tag{33}$$

$$\sum_{\ell>j} d_{\ell,t} \le \overline{P} v^d_{j,t}, \qquad h_{j,t+1} \le (E/J)(1 - v^d_{j,t}). \tag{34}$$

Using a deeper segment therefore requires the shallower segment to end full when charging or empty when discharging. This chooses a deterministic shallow- first representative, including otherwise indifferent charging allocations. Segment actions have the same information ownership as aggregate battery power.

At a declared fresh study start, initialize

$$h_{j,0} = \text{clip}\left(e_0 - (j-1)E/J,\ 0,\ E/J\right). \tag{35}$$

Afterwards the segment vector must be carried with actual execution; an arbitrary residual solve cannot reconstruct it from aggregate SOC. Previous throughput or aging requires explicit memory. Xu mode currently rejects nonzero self-discharge. It remains a finite-segment approximation and omits calendar, temperature and current-rate aging, capacity-fade feedback and unknown pre-study cycle history.

6 Submission, publication and reoptimization workflow

Before each active auction submission, the GME replay refreshes the latest available PV forecast over unexecuted intervals, projects a boundary if necessary, and solves the residual planning problem. Previously submitted quantities, including still-unpriced ones, are fixed. At submission, only the current auction's node-shared action is extracted. Later planned auction quantities are not booked and can change.

At publication, known prices are updated, and actual contractual revenue $\pi^{\text{real}}_{a,t} q^{\text{sub}}_{a,t}$ is entered exactly once in the ledger. Publication does not reopen a market. The current GME experiment forces `reoptimize_physical_controls=false`; it uses four auction solves for four active auctions, not an extra set of publication solves or physical optimizations. Physical feedback is algebraic under this retained-plan policy. The optional continuous MPC policy adds interval-start solves as described below. Synthetic replay can separately enable interval-control reoptimization, so its solve count need not be four.

At the next auction, closed commitments are carried, remaining open opportunities are reconsidered, and past executed state/aging remain fixed. In the actual-price oracle all future auction prices are already supplied as a benchmark, so price publication does not remove price uncertainty there. In a forecast-based route, known results condition subsequent price distributions without changing the commitments that produced them.

7 Optional real-time economic MPC

The default remains `track_committed_grid`. Four additional continuous execution selectors combine deterministic/ensemble PV with price-blind/oracle imbalance treatment: `deterministic_mpc`, `deterministic_mpc_price_oracle`, `ensemble_mpc`, and `ensemble_mpc_price_oracle`. They add physical control solves at each quarter-hour start, over the remainder of that local delivery day.

Auction submission, publication, fees and eligibility are unchanged. In particular, planned future orders are not fixed positions in this lower-level model.

For each MPC horizon, Q_t^{sub} contains only quantities already submitted, including unpriced commitments. Battery powers c_t, d_t, mode binaries, energy e_t and a utilization fraction $a_t \in [0,1]$ are common across *all* scenarios for the *entire* horizon. The additional equations are

$$i_{s,t} = a_t V_{s,t}, \qquad g_{s,t} = i_{s,t} + d_t - c_t, \tag{36}$$

$$\delta_{s,t} = g_{s,t}\Delta_t - Q_t^{\text{sub}}, \qquad L = \sum_s w_s \sum_t |\delta_{s,t}|. \tag{37}$$

The shared battery transition (13), power exclusivity, efficiencies, SOC box and POI bounds apply. MPC and auction builders reuse common battery and degradation helpers. Scenario-specific battery recourse and arbitrary scenario-specific curtailment are absent. The first utilization fraction is applied to realized available PV by the execution environment after issuance.

Price-blind MPC minimizes

$$\kappa L + D^{\text{plan}} + C^{\text{import}} + C^{\text{terminal}}, \qquad \kappa = 500 \text{ EUR/MWh by default.} \tag{38}$$

There is no independent curtailment penalty. The default terminal cost is $50|e_T - 0.5E|$ EUR, with energy in MWh; no-value and configurable linear-value alternatives are supported. Separately configured final-study hard targets and explicit horizon-end minima remain constraints. A hard 50% equality is not imposed on ordinary operating midnights. Planning terminal costs never become real cash charges.

The imbalance-price oracle first minimizes L with the same physical and hard terminal constraints. After solver certification within the recorded tolerance, it maximizes expected *signed* settlement minus degradation, import and terminal costs, subject to

$$L \le L_{\min}(1 + r) + b. \tag{39}$$

The relative and absolute budgets r, b default to zero. Actual future Terna prices enter only an explicitly marked noncausal input object; price-blind MPC does not accept it. Single-price cash flow is $\lambda_t \delta_{s,t}$, including negative prices, rather than $-|\lambda_t||\delta_{s,t}|$. Distinct directional curves retain sign exclusivity. With a single curve, the nonnegative split is exact in the minimum-volume stage and conservatively bounds true absolute volume in the economic stage; reporting reconstructs physical volume directly.

One configurable ten-second wall budget covers building and both solver stages. Stage objectives, bounds, gaps and actual additional imbalance are logged. An uncertified first-stage incumbent cannot authorize an economic second stage; it can supply a validated price-blind action. If economic refinement has no usable solution, a validated first-stage action is retained. No valid action invokes the explicitly logged existing tracking fallback. Solver tolerances remain relative MIP gap 10^{-4} with no configured absolute gap.

Information comprises current observed energy and degradation memory, completed interval PV, submitted commitments, issued commands and eligible forecast vintages. By default current and future realized PV are absent from MPC inputs. The explicit `realtime_mpc.current_pv_information` option can instead disclose the realized current-interval average before solving. It replaces $V_{s,0}$ by that same measured/effective PV value in every member while all later intervals remain forecast-only. This is a retrospective current-PV oracle, separate from the imbalance-price oracle; both disclosures are recorded. The ensemble uses the preceding delivery day's frozen 00 UTC weather members and a latest eligible deterministic member, uniformly weighted by default; explicit normalized weights and a weather-only sensitivity are supported. The weights are not calibrated probability claims. The revised two-day Farm 1 examples use the selected `direct_tcn_02`

bundle for deterministic/ensemble conversion, initialization, measurement fallbacks and the common auction-source forecasts, without fitting. Its training period is 2025; its January–April 2026 selection and October 2026 artifact registration make January reuse retrospective. Checksums and this hindsight selection are retained. The initial pvlib/current- forecast comparison is preserved separately; the operational default remains pvlib with replacement disabled.

Only the first action executes, through `follow_battery_plan` protection without economic grid-tracking replacement. Interval-average realized PV can change controls only through declared physical protection. Full plans, first commands and overrides are saved separately. Auction events cannot retrospectively replace an unfinished interval's command.

MPC auction-boundary projection follows the last completed open-loop plan from the latest actual state, preserving an outstanding action and using the latest eligible *deterministic* PV forecast at the auction cutoff, even for ensemble MPC. It does not solve hypothetical future MPCs or choose a weather member. Missing plan coverage invokes logged forecast-only tracking/protection. In current-PV-known mode the locked unfinished interval uses the scalar already disclosed at issuance, while subsequent intervals retain deterministic forecast PV; the shadow rollout never retrieves future actual production. The result, `projected_soc_mpc_deterministic`, never overwrites actual state. Pure battery-power commands can make projected energy independent of PV until protection acts. Realized noon/midnight projection errors are recorded; actual carryover is not reset to tomorrow's earlier assumed state.

The primary comparison freezes one existing integrated deterministic auction ledger and replays identical timed submissions with each controller. Auction revenue and fees must match. The secondary full continuous comparison lets actual SOC and MPC projections change later auction orders. Their interpretations remain separate. The auction-price oracle is unchanged and distinct from the new lower-level imbalance-price oracle. Tracking's interval-average PV feedback advantage is disclosed in causal mode; the current-PV-known variant supplies the same interval-average production to MPC. An optional causal tracking diagnostic is available. Detailed configuration, metrics, limitations and commands are in `docs/realtime_economic_mpc.md`.

8 Physical feedback with realized PV

8.1 Feasible interval power range

The current policy is `track_committed_grid`. For interval t its target is

$$g_t^\star = Q_t^{\text{committed}}/\Delta_t. \tag{40}$$

The controller uses effective realized interval-average PV $\widetilde{V}_t$; this is an idealized interval simulator, not a subsecond telemetry/control model. It does not disclose this final average to the earlier auction optimizer. Given actual initial energy e_t and retained energy $\bar{e}_t = \rho_t e_t$, the feasible charging/discharging maxima are

$$C_t = \min\left\{\overline{P}, \frac{[E^{\max} - \bar{e}_t]_+}{\eta_c \Delta_t}\right\}, \tag{41}$$

$$D_t = \min\left\{\overline{P}, \frac{\eta_d \left[\bar{e}_t - E^{\min}\right]_+}{\Delta_t}\right\}. \tag{42}$$

First clip the planned charge/discharge commands to these maxima, obtaining c_t^0, d_t^0. Simultaneous positive requested charging and discharging after protection is rejected. Initially curtail only available PV: $u_t^0 = \min\{\widetilde{V}_t, \left[u_t^{\text{plan}}\right]_+\}$.

8.2 Grid-tracking policy

Release scheduled curtailment if it can help meet a deficit:

$$g_t^0 = \widetilde{V}_t - u_t^0 + d_t^0 - c_t^0, \tag{43}$$

$$\bar{u}_t = u_t^0 - \min\{u_t^0, \left[g_t^\star - g_t^0\right]_+\}. \tag{44}$$

The desired net battery dispatch and its feasible projection are

$$b_t^{\text{req}} = g_t^\star - \widetilde{V}_t + \bar{u}_t, \tag{45}$$

$$\tilde{b}_t = \text{clip}(b_t^{\text{req}}, -C_t, D_t). \tag{46}$$

Finally curtail any remaining surplus that battery adjustment cannot absorb:

$$\tilde{u}_t = \bar{u}_t + \min\{\widetilde{V}_t - \bar{u}_t, \left[\widetilde{V}_t - \bar{u}_t + \tilde{b}_t - g_t^\star\right]_+\}. \tag{47}$$

The controller then recovers curtailed PV using the available battery headroom, without changing grid exchange. It reduces discharge before increasing charge:

$$r_t = \min\{\tilde{u}_t, \left[\tilde{b}_t + C_t\right]_+\}, \tag{48}$$

$$b_t = \tilde{b}_t - r_t, \qquad u_t = \tilde{u}_t - r_t, \tag{49}$$

$$c_t = \left[-b_t\right]_+, \qquad d_t = \left[b_t\right]_+. \tag{50}$$

This is an unconditional feedback action, rather than a new optimization; its SOC and degradation effects carry forward to later intervals. The saved `curtailment_recovered_mw` measures the curtailment reduced in this feedback step, whose starting value depends on the active plan. It can include avoided discharge as well as extra charging and depends on the plan; it is not itself a direct measure of energy newly charged into the battery. Since the first-member plan is used by the controller, reconstruction of that member's curtailment after envelope reduction is necessary for consistent execution. Thus excess PV first decreases discharge, then increases charging as needed and feasible, then is curtailed. A PV shortfall first releases planned curtailment, then reduces charging or increases discharge within the actual SOC box. The remaining tracking error is preserved if battery power/energy cannot supply it.

POI protection follows. If net import exceeds its rating, charging is reduced. If export exceeds its rating, discharge is reduced first and remaining excess PV is curtailed. Grid power is recomputed consistently with the adjusted actions. Actual energy then advances using (13) with these executed powers, and bounds are checked before accepting the new state. Both throughput/Xu memory and cumulative financial wear are advanced from executed actions, not from the old auction plan. Protection is mandatory under either execution policy; the alternative `follow_battery_plan` skips grid-tracking adjustment.

There is no controller constraint forcing the actual terminal energy to the auction plan's target. Forecast errors and grid tracking can cause an ending-SOC difference even if every optimization imposed (15). This is visible in the current terminal-50% study and is not equivalent to guaranteeing 50% realized SOC. Any hard actual-terminal policy would be an additional controller design and could create different imbalances/cash flows.

8.3 PV source allocation for reporting

For the executed trajectory, use the explicit PV-first accounting convention

$$P_t^{\text{PV}\to\text{B}} = \min\{\widetilde{V}_t - u_t, c_t\}, \tag{51}$$

$$P_t^{\text{PV}\to\text{G}} = \widetilde{V}_t - u_t - P_t^{\text{PV}\to\text{B}}, \tag{52}$$

$$P_t^{\text{PV}\to\text{curt}} = u_t. \tag{53}$$

They sum to $\widetilde{V}_t$ in every interval. Grid-supplied charging and battery discharge are not counted as PV production. These are post-processing flows; there are no separate source-attribution decision variables in the MILP.

9 Settlement and expected-versus-realized accounting

Actual net deviation is retained at the native quarter-hour:

$$\delta_t^{\text{real}} = g_t^{\text{real}} \Delta_t - Q_t^{\text{committed}}. \tag{54}$$

Positive means more net export than promised. With the selected single Terna NORD settlement price λ_t, the modeled cash flow is

$$S_t = \lambda_t \delta_t^{\text{real}}. \tag{55}$$

A positive price therefore does not mean every imbalance is penalized: positive excess export is credited and a negative deviation is debited. Negative prices are allowed and reverse these cash-flow signs. The general settlement helper also accepts two directional curves:

$$S_t = \lambda_t^{+} \left[\delta_t^{\text{real}}\right]_{+} - \lambda_t^{-} \left[-\delta_t^{\text{real}}\right]_{+}. \tag{56}$$

This documents the implemented study accounting, not every possible Italian participant-specific settlement item or regulatory regime. The workbook adapter selects the declared NORD macrozone and the imbalance-price column, retaining other imported fields as provenance. Product starts and ends must match the execution grid exactly. Prices are neither hourly-averaged nor filled across gaps; missing enabled settlement prices are an error.

Actual daily auction cash flow and settled operating profit are

$$A^{\text{real}} = \sum_{a,t} \pi_{a,t}^{\text{real}} q_{a,t}^{\text{sub}}, \tag{57}$$

$$J^{\text{real}} = A^{\text{real}} + \sum_t S_t - D^{\text{executed}} - f \sum_t \left[-g_t^{\text{real}}\right]_{+} \Delta_t - C^{\text{trading,real}}. \tag{58}$$

The stochastic-ensemble model deterrence penalty is not an actual settlement charge and is not subtracted from realized profit. Legacy non-ensemble runs with optimized imbalance retain their existing configured realized penalty. If settlement is unavailable, profit before settlement is reported and fully settled profit remains unknown in the settlement helper and compatible independent-day outputs. The continuous comparison runner requires complete actual settlement prices and rejects a run when settlement is disabled or incomplete. The continuous ledger additionally decomposes realized auction cash flow as

$$A^{\text{real}} = \sum_t \pi_{DA,t}^{\text{real}} \sum_a q_{a,t}^{\text{sub}} + \sum_{a \neq DA,t} (\pi_{a,t}^{\text{real}} - \pi_{DA,t}^{\text{real}}) q_{a,t}^{\text{sub}}. \tag{59}$$

Each signed quarter-hour quantity is in MWh, including the allocated shares of native hourly products; no second duration factor is applied. If the DA reference curve is absent, both components are unavailable rather than filled. Offsetting volume is the sum across products of the smaller of gross purchases and gross sales. Matched PV-only continuous replays remove battery capacity and power while retaining the forecast and market policy. Their cash-profit difference does not assign a terminal value to remaining stored energy. Only actually submitted guaranteed transactions enter $C^{\text{trading,real}}$; future plans that are never submitted contribute no realized fees. The ledger retains transaction-month volume, per-trade charges and tariff versions. An independent Python calculation reconciles those charges and the final monthly state. Expected and realized transaction fees are reported separately from gross auction revenue. In `realized_profit_breakdown.png`,

trading-fee EUR and gross MWh occupy the former import-fee slot; nonzero physical grid fees appear as their own additional bar.

Let J^{init} and A^{init} be the initial auction solve's expected profit and auction revenue. Saved mismatches are

$$\Delta J = J^{\text{real}} - J^{\text{init}}, \qquad \Delta A = A^{\text{real}} - A^{\text{init}}, \qquad r_J = 100\frac{\Delta J}{|J^{\text{init}}|}. \tag{60}$$

The percentage is undefined/`null` when its expectation is zero; unsettled profit comparisons remain unknown. Differences include later replanning and physical execution effects, not only price-forecast error.

The replay also evaluates each stored auction snapshot as an open-loop diagnostic against realized PV and the same settlement prices. Its planned battery commands are protected physically. Deviations are measured against that snapshot's planned grid schedule (which can include assumed future trades) or the final contract, as labeled in the recorded comparisons. It is not another auction optimization. Because IDA3 covers a shorter residual horizon, a common-horizon comparison is also saved; unaligned whole-horizon totals should not be compared as if their delivery intervals were identical.

10 Value of reoptimization through frozen-plan replay

The optional frozen-plan comparison asks what would happen if planning stopped after one auction decision but physical feedback continued. This differs from the older open-loop imbalance diagnostic, which follows planned battery commands without grid-tracking feedback and does not reconstruct an alternative complete auction ledger and net profit.

Let k be an active auction other than the final active auction. At its cutoff, retain the entire deterministic future-trade vector from that solve. Define

$$q_{a,t}^{[k]} = \begin{cases} q_{a,t}^{\text{submitted,ref}}, & a \le k, \\ q_{a,t}^{\text{plan},k}, & a > k, \end{cases} \qquad Q_t^{[k]} = \sum_a q_{a,t}^{[k]}. \tag{61}$$

The already submitted trades are fixed even if their prices are unpublished. Future orders are the frozen snapshot's orders, not the later realized orders from the reoptimized strategy. They are submitted at their original gates and priced at actual clearing prices. Their eligibility must ensure submission precedes the relevant delivery interval. Thus their cumulative position is a valid delivery-time tracking target without prematurely revealing new prices or forecasts to the frozen plan.

Executed history before the frozen residual boundary remains identical to the reference trajectory. The suffix starts from the actual carried energy, throughput, incurred wear and, for Xu mode, segment-memory vector. It is not reset to the snapshot's forecast SOC, including when the cutoff projects a not-yet-observed boundary. At physical delivery that boundary's actual state is carried forward without changing the frozen decisions. The same configured execution policy uses the frozen battery/curtailment commands, target $Q_t^{[k]}/\Delta_t$, and the same effective realized PV:

$$(c_t^{[k]}, d_t^{[k]}, u_t^{[k]}, e_{t+1}^{[k]}) = \mathcal{F}(e_t^{[k]}, \widetilde{V}_t, Q_t^{[k]}/\Delta_t, c_t^{\text{plan,k}}, d_t^{\text{plan,k}}, u_t^{\text{plan,k}}). \tag{62}$$

Here $\mathcal{F}$ is the grid-feedback/protection policy described above. The current study uses grid tracking; the alternative execution policy, if selected, is shared with the reference. No later optimization or weather-forecast update is applied. The algebraic replay adds zero optimization solves.

Using the same signed settlement convention and actual dated transaction-fee accounting as the reference, the full-day frozen profit is

$$J_{\text{frozen}}^{[k]} = \sum_{a,t} \pi_{a,t}^{\text{real}} q_{a,t}^{[k]} + \sum_t S_t^{[k]} - D^{[k]} - C_{\text{import}}^{[k]} - C_{\text{trading}}^{[k]}. \tag{63}$$

Previously executed prefix costs are preserved. Future wear is recomputed from the counterfactual's executed commands; gross fees are recomputed from its own trade ledger and the same initial operator-volume state, including month transitions and tariff bands. Reference future trades or fees are never borrowed after the freeze point.

The value KPI is

$$G_k = J^{\text{real}}_{\text{fully reoptimized}} - J^{[k]}_{\text{frozen}}. \tag{64}$$

Both sides cover the entire delivery day, so differing residual lengths do not create incomparable profit totals. Positive values favor later reoptimization; negative values are retained. Each G_k measures all later replanning after k: the values are not additive and need not be monotonic. This comparison includes new forecasts, revised orders and state-aware planning rather than isolating a causal effect of forecast accuracy. Final stored-energy differences are reported; no terminal salvage price is assumed. If actual settlement is unavailable, settled profit and G_k remain unknown.

The independent YAML section is `reoptimization_baseline`. Its options are `enabled`, `freeze_after_auctions`, and `plots`. An absent section disables the feature; the comparison study enables it. A null auction selection evaluates every active pre-final auction, while a list chooses a subset. The last active auction is always excluded, cancelled auctions produce no baseline, and an empty selection produces no comparisons. The plot switch also respects the global plot setting. Snapshot replay requires deterministic plans; it rejects using one selected leaf of a contingent plan as if all of its future branches were already known.

The simulator is `src/simulation/frozen_plans.py`. Saved dispatch-plan rows include every `planned_trade_<auction>_mwh` and their planned net position. Full-day profits, gains and states are saved in the frozen-plan summary CSV/JSON, with separate actual counterfactual dispatch and trade ledgers. Individual gains also enter the scalar KPI table, and `reoptimization_value.png` compares frozen profits with the reference. The isolated January configuration is `configs/full_experiments/gme_oracle_2025-01-01_farm910_reoptimization_value.yaml`.

11 Perfect-information schedules and the PV-only baseline

11.1 PV+BESS versus the feedback controller

After replay, one additional full-day problem uses $V_t = \widetilde{V}_t$ and all actual auction-price curves, with a single scenario and all auction orders jointly free subject to their native products, eligibility, POI and physical constraints. It retains the BESS ratings, efficiencies, SOC box, degradation, initial energy and configured terminal target. Call its achieved solver objective $J^{\text{PI}}_{\text{PV+BESS}}$.

This is deliberately a hindsight plan rather than a causally executable policy. It is compared with the feedback controller's actual dispatch, curtailment and SOC, and its objective is saved as the solver's `ObjVal`. Four missing measurement intervals, if permitted by the selected run, are handled through that run's explicit fallback; diagnostics record their count and identities. The schedule figure shows both trajectories and their battery-power difference through the final day boundary. It does not assert that feedback should reproduce the hindsight trajectory or terminal energy.

11.2 PV-only asset counterfactual

When `experiment_outputs.pv_only_baseline: true`, a second independent solve uses the same optimizer builder with

$$E = 0, \quad \overline{P} = 0, \quad E^{\min} = E^{\max} = 0, \quad e_0 = e_T = 0, \quad D = 0. \tag{65}$$

Battery powers and energy variables are consequently fixed to zero; the code reuses one physical/trading formulation instead of implementing a second, potentially inconsistent PV-only model.

Unused variables can be removed by solver presolve. PV injection/curtailment and auction trades remain free subject to the same effective realized PV, native products, active auctions and POI. Call its achieved objective $J^{\mathrm{PI}}_{\mathrm{PV}}$.

The incremental operating-value KPI is

$$\Delta J^{\mathrm{PI}}_{\mathrm{BESS}} = J^{\mathrm{PI}}_{\mathrm{PV+BESS}} - J^{\mathrm{PI}}_{\mathrm{PV}}. \tag{66}$$

The BESS side includes its own modeled degradation; the PV-only side has no battery wear. Both sides retain permitted cross-auction spread trading, so market-trading revenue common to both is not credited to battery ownership. The KPI does not subtract realized feedback wear from a different ideal schedule. Each counterfactual reports its own objective components. Directly subtracting a PV-only hindsight objective from the realized PV+BESS feedback profit would mix information and execution policies and is not the asset-value KPI.

This comparison is one-day operating value before capital expenditure, fixed operating costs, financing and taxes. A free terminal target can also use initial stored energy or leave different inventory; the selected 50% initial/terminal hindsight comparison avoids that particular inventory difference for the BESS model. Realized feedback may still finish at a different SOC. No residual-energy valuation, annualization, payback, NPV or investment optimization is implemented. Those need a representative horizon and additional cost/cash-flow assumptions.

Setting the switch to `false` skips the PV-only solve and its model, JSON, CSV and figure. Incremental-value metrics become `null` and an explicit baseline-enabled KPI is false. The perfect-information PV+BESS-versus-feedback comparison still runs. Four active auctions therefore produce four replay solves plus one PV+BESS hindsight solve and, when enabled, one PV-only solve. Counterfactual solve diagnostics are separate from replay solve counts.

12 Configurable participation and matched-day profile comparison

Participation is a choice of a nonempty subset P of the chronologically ordered auction set $\mathcal{A} = \{\mathrm{DA}, \mathrm{IDA1}, \mathrm{IDA2}, \mathrm{IDA3}\}$. The single-study configuration is `gme_oracle_experiment.participating_auctions`. Omission or null selects every configured auction dataset. Gate order remains unchanged; unselected auctions have no price-data requirement, trade variables, submission event/solve or fees. Their orders are absent rather than independent variables fixed after a solve. The active set on day d also removes documented/configured cancellations. If every selected auction is documented/configured as cancelled, the profile retains its full delivery day with zero orders, auction/control solves and auction fees. The zero-position feedback policy stores PV within the SOC box and curtails remaining surplus; executed wear and settlement remain in actual profit. There is no initial auction objective (its expectation KPIs are null), and no auction plan enforces the configured terminal target on that day. Missing or partial price data without cancellation evidence still fail.

Four independently optional auctions produce

$$|\{P \subseteq \mathcal{A} : P \neq \varnothing\}| = 2^4 - 1 = 15. \tag{67}$$

If DA is mandatory and the three IDA auctions remain optional, there are instead

$$|\{\{\mathrm{DA}\} \cup S : S \subseteq \{\mathrm{IDA1}, \mathrm{IDA2}, \mathrm{IDA3}\}\}| = 2^3 = 8. \tag{68}$$

Both families and explicitly listed profiles are supported. For example, $P = \{\mathrm{IDA1}, \mathrm{IDA2}\}$ and $P = \{\mathrm{IDA3}\}$ are valid without DA. All physical, native-product, degradation and settlement constraints continue to apply to the same full delivery day.

If the first selected gate occurs after delivery begins, its prefix uses zero contracted grid delivery and grid feedback with zero planned battery commands. The controller may store PV within the energy box and curtail remaining surplus; it does not earn uncontracted export revenue in this prefix. Actual energy and wear carry to the first selected solve, including the existing unfinished-interval state projection. No extra auction optimization is inserted before the first selected decision. Prefix executed/projected costs are subtracted from the first residual objective when forming an initial whole-day expected profit; the residual model objective is separately retained. Actual whole-day profit already includes all executed prefix costs through the ordinary financial accounting.

12.1 Sequential current-auction-only baseline

At the decision for auction a, integrated planning includes the open auctions $\mathcal{A}_a$. The configurable sequential baseline replaces this set by $\{a\}$. Let $Q_t^{<a}$ denote all previously submitted, fixed quantities, including unpriced commitments. Its only free market quantity is $q_{a,t}$:

$$Q_t = Q_t^{<a} + q_{a,t}, \qquad \max \sum_t \lambda_{a,t} q_{a,t} - C_{\text{deg}} - C_{\text{import}} - C_{\text{fees}}. \tag{69}$$

The existing physical balances link Q_t to grid delivery. The full remaining delivery horizon retains charge/discharge, curtailment, SOC transitions, native product equalities, POI limits and the configured terminal target. No later auction trade or revenue appears. Sunk earlier cash flow does not affect the choice. Each next gate receives updated PV, actual carried battery state and immutable prior submissions. Under the price-oracle convention, the current auction's actual clearing curve is supplied to that solve.

`gme_oracle_experiment.auction_planning_mode` selects `integrated` (default) or `current_auction_only`. `participation_study.sequential_all_auctions_baseline` optionally adds one all-auction sequential result beside the integrated participation profiles. The January study refresh comprises 465 integrated cases and 31 sequential cases. Matched daily realized profits, their paired differences, model sizes, forecast inputs and fallback decisions are saved. A later infeasible sequential solve retains prior physical commands and submits zero new trades; it has no previously planned future order to submit.

12.2 Operational fallback after an infeasible auction decision

Exact planned delivery does not guarantee feasible recourse after forecast updates. Limited remaining energy, power and trading capacity may conflict with the configured terminal state or remaining delivery obligations. The physical controller can continue within SOC/power limits by accepting an imbalance even when the planning model is infeasible. A pre-noon delivery shortfall is executed and settled under the retained policy; those closed intervals are excluded from the IDA3 MILP and cannot make it infeasible.

The configurable policy `execution.infeasible_auction_policy` defaults to `error`. The full-January study uses `previous_plan`. When and only when the solver confirms infeasibility, this policy uses the last feasible deterministic snapshot for the current auction's previously unsubmitted orders and remaining requested physical commands:

$$q_{a,t}^{\text{submit}} = q_{a,t}^{\text{last feasible}}, \qquad u_t^{\text{requested}} = u_t^{\text{last feasible}}. \tag{70}$$

The orders are submitted at their actual gate and priced at actual clearing prices with ordinary fee accounting; existing commitments remain fixed. The same feedback controller executes against actual PV, respects the physical limits, and settles unavoidable deviations ex post. This is a retained plan, not a new feasible optimum under the updated forecast. No recovery optimization or constraint relaxation is introduced. Subsequent selected decisions still attempt their normal

reoptimization. Missing prior plans, nondeterministic snapshots and other solver failures remain errors.

Failed decision/forecast diagnostics, reused plan sources, submitted quantities and attempt counts are saved separately from successful solves; optional IIS reports use `execution.save_infeasibility_diagnostics`. Summary tables retain fallback-day and failed-decision counts rather than omitting those dates from the comparison. The physical trajectory can finish away from the terminal SOC target of its retained plan, just as under ordinary forecast-error feedback.

The `experiment-participation` command runs configured profiles on a common set of delivery days $\mathcal{D}$. The ranking metric is

$$\overline{J}(P) = \frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} J^{\text{real}}(P, d), \qquad P^{\star}_{\mathcal{D}} \in \arg\max_{P} \overline{J}(P). \tag{71}$$

It is an empirical ranking on $\mathcal{D}$, not an established optimum for unseen days. Selecting a different winning profile after each observed day would evaluate $|\mathcal{D}|^{-1} \sum_d \max_P J^{\text{real}}(P, d)$ and would use additional hindsight; the saved daily-winner table is diagnostic and is not the fixed-profile ranking.

Every profile must complete exactly the same requested dates with fully priced actual settlement. Missing or invalid inputs fail the batch; they do not silently change a profile's averaging sample. Battery and operator fee initial state are identical across profiles and reset independently each study day. This is not a continuous multi-day carry of SOC or monthly traded volume. The initialization weather cutoff is common before the earliest candidate gate and delivery start. Later forecast updates occur only at selected auction decisions. Delivery grids, initial PV forecasts, effective realized PV (including configured fallback intervals) and shared actual auction-price curves are fingerprinted and checked for equality across profiles. The market adapter resolves native auction-product durations for each date. Actual price snapshots, model inputs and all daily decisions remain in the per-profile run artifacts.

Summary outputs retain average, total, minimum, maximum and median daily profit, 10th/90th percentiles, daily population standard deviation, negative-profit and zero-active-auction day counts, daily wins including configured tolerance ties, paired differences to the largest selected profile, mean SOC/energy, curtailment, wear, transaction fees and active auction/solve counts. Final stored energy is reported without salvage valuation. The current configured terminal SOC remains a planning constraint rather than a guarantee on feedback's final state, so cash profit and ending inventory must be assessed together. Fee and inventory reset assumptions also matter when extrapolating to a longer investment horizon.

The comparison figure places a separate gold perfect-foresight reference above the ranked profiles. Its mean and daily standard deviation use the full-day perfect-information objective values (Gurobi `ObjVal`) on the identical dates, effective realized PV, assets, costs and configured terminal SOC, with every configured auction that remains available. This reference includes markets outside a restricted candidate/profile family. Existing all-auction benchmark outputs are reused; if that profile is absent, one additional perfect-information solve per day supplies the reference without another auction replay. Its daily objectives and summary are retained separately, along with objective-minus-profit differences in the profile tables. This is a model-objective reference rather than an additional feedback participation rank: actual imbalance settlements and realized terminal inventory can differ between the accounting cases. `participation_study.perfect_foresight_reference` enables this feature by default and can be set to false.

The batch YAML section `participation_study` controls candidate/required auctions, explicit profile lists, common dates, output paths, per-profile figures/models, summary plotting and profit-tie tolerance. The supplied example is `configs/full_experiments/gme_participation_2025-01-01_03.yaml`. The full-month independent-day configuration is `configs/full_experiments/gme_participation_2025-01.yaml`, with 31 common delivery days and 465 profile/day cases. Fully cancelled IDA3-only cases remain in the averaging sample rather than changing that

profile's denominator. The driver is `src/simulation/participation_study.py`, while `src/participation.py` validates/enumerates subsets. Summary CSV/JSON/plots are stored apart from each individual replay under its profile/day directory. Optional `participation_study.resume` verifies the same effective configuration, completed child manifests, input fingerprints and saved profits before reusing checkpoints. Incomplete children are retained separately and rerun; changed configurations are rejected. After independent solution validation, tolerated negative floating-point command roundoff is floored to zero before the physical controller's strict nonnegative-input checks.

13 Continuous operation with daily optimization horizons

13.1 Delivery books and the chronological event sequence

The separate `experiment-continuous` workflow selects inclusive local delivery dates $d \in \mathcal{D} = \{d_0, \ldots, d_F\}$ through `continuous_optimization.start_date` and `end_date`. Each day has its own physical grid $\mathcal{T}_d$, native auction products, actual price curves and auction calendar. Commitments and revisable plans are keyed by delivery day as well as auction and interval, for example (d, a, t); the same auction name on another delivery day is a different market book. This is continuous chronological execution with daily planning horizons; it is not a single multi-day MILP, and a later-day solve cannot revise an earlier delivery book.

All events are merged into one chronological sequence. On an ordinary operating day d, the replay submits IDA3 orders for d, then DA, IDA1 and IDA2 orders for $d+1$ while executing delivery d. Submissions use the configured decision buffer before the local gate; price publications book already submitted orders without triggering another optimization. IDA3 variables remain restricted to noon–midnight of their delivery day. Native DA-hour equalities and quarter-hour IDA/physical controls are unchanged. Cancelled auctions produce no order, variables or solve, and unexplained missing inputs fail preflight.

This is a chronological sequence of *daily* MILPs, not a simultaneous month-long or two-day trading MILP. An optimization for $d+1$ cannot change current-day dispatch or commitments for d. Conversely, current-day IDA3 cannot reopen submitted next-day DA quantities. The current-day commands and book enter next-day planning through the projected boundary state below.

13.2 Forecast-based projection of the next initial state

For the default retained-plan controller, at a decision cutoff c_k, let x_k^{obs} be the most recent observed physical state at boundary $\tau_{i_k} \le c_k$. This state includes energy, cumulative executed throughput/degradation and, when relevant, depth-model segment history. The current continuous controller requires zero configured telemetry latency, treating an interval's actual state as observable at its end; preflight rejects a nonzero `source.telemetry_latency_minutes`. This restriction is independent of the weather forecast's availability lag. An unfinished interval's already issued command is retained. Let $\bar{u}_i$ be the retained current-day charge, discharge and PV-curtailment commands and Q_i^{sub} the cumulative submitted grid position. For a pre-delivery DA, IDA1 or IDA2 solve for d, the planning boundary $b(d,a)$ is midnight at the start of d; for IDA3 it is noon on d. Define a shadow state using only the weather/PV vintage available at c_k:

$$\widehat{x}_{i_k|k} = x_k^{\mathrm{obs}}, \tag{72}$$

$$\widehat{x}_{i+1|k} = \Phi\Big(\widehat{x}_{i|k}, \bar{u}_i, Q_i^{\mathrm{sub}}/\Delta_i, \widehat{P}_{i|c_k}\Big), \qquad i_k \le i < b(d,a). \tag{73}$$

Φ is the same grid-tracking and physical-protection controller used for actual execution. Thus the forecast projection respects SOC, battery power and POI limits while preserving the old commands and submitted grid target. It uses no new optimization for the locked prefix and no future measured

PV. Forecast eligibility uses the UTC publication/availability clock at c_k, including the configured six-hour lag, independently of the later projected delivery boundary.

The daily MILP fixes its initial state to $x_{d,b} = \widehat{x}_{b(d,a)|k}$. At a later gate this projection is recomputed from the newer actual state and latest eligible forecast. Meanwhile actual execution advances separately:

$$x_{i+1}^{\text{real}} = \Phi\Big(x_i^{\text{real}}, \bar{u}_i, Q_i^{\text{sub}}/\Delta_i, P_i^{\text{real}}\Big). \tag{74}$$

Neither solving nor revising a shadow projection overwrites actual energy or aging state. The realized midnight state may differ from the previously projected state; later solves carry the actual state as it becomes available. MPC instead uses the deterministic open-loop procedure in Section 7, supplying one state to the same auction MILP. Saved boundary records retain prediction, actual state and signed error $E_b^{\text{real}} - \widehat{E}_{b|k}$. Midnight projections are distinguished from IDA3 noon projections in diagnostics and plots.

13.3 Daily free decisions, initial condition and final target

For integrated planning, the free trade quantities are those of the current and remaining active auctions *for that delivery day*. For sequential `current_auction_only` planning, only the current auction has free trade variables and expected trading revenue. Previously submitted quantities for the target book, including unpublished pending orders, remain fixed in $Q_{d,t}^{\text{fix}}$ in both modes. Their cash values remain external to the new decision objective. Physical controls and SOC over the target day's remaining eligible delivery horizon remain free under the existing balances, costs and asset constraints. Trades in other delivery books are not decision variables in this MILP. Its objective is the daily residual objective (22); projected costs already locked before its planning boundary are not new trading opportunities.

For the supplied January configuration, $d_0 = 1$ January 2025 and $d_F = 31$ January 2025. The supplied boundary is

$$E_{d_0,0}^{\text{real}} = 0.90\, E^{\text{nominal}}. \tag{75}$$

Trading starts with the 31 December DA/IDA1/IDA2 events for delivery on 1 January. Their initial midnight state is this supplied assumption; it is not inferred from unavailable 31 December physical operation. Eligible 31 December forecasts and the actual 1 January delivery-price curves are still required. After the first midnight, there is no daily SOC reset.

The configured terminal constraint is applied only when planning the final delivery day and only when the target is non-null:

$$e_{d_F,|\mathcal{T}_{d_F}|} = E^{\text{nominal}}\, s_F/100, \qquad s_F = 50 \quad \text{(SOC percentage in the January example)}. \tag{76}$$

Under the default retained-plan controller, intermediate midnight states have no equality target or energy salvage price. Optional MPC instead uses its configured inventory treatment from Section 7. They remain inside their SOC box and, when configured, satisfy the additional planned minimum in (16). The January 2025 template leaves that minimum disabled; the paired January 2026 studies use 40%. Actual terminal energy remains an output of feedback execution and need not equal the planned target. The last delivery interval ends at local midnight on 1 February. The final day's IDA3 and all other auctions delivering on 31 January are included. On 31 January the DA, IDA1 and IDA2 auctions for 1 February delivery are outside the study and are not submitted or counted in January profit.

13.4 Independent trajectories, accounting and restart

The default experiment executes three independent chronologies: integrated planning, sequential current-auction-only planning and an integrated perfect-PV reference with the same daily horizons.

All start from the same supplied state and use matched actual price, settlement, effective measured PV and asset inputs, but each carries its own decisions and realized state. The reference replaces forecast PV by effective realized PV, including in shadow projections. It does not introduce future delivery-day trade variables. Its reported performance is continuous settled operating profit, not the sum of overlapping auction objective values. The two trading modes and this reference are independently configurable.

The reference defaults to one optimization per delivery day, at DA (or the first active auction if DA is cancelled). The flag `continuous_optimization.perfect_pv_reoptimize_at_each_auction` defaults to `false`; setting it to `true` restores optimization at every active gate. With the default, all later auction quantities are read from the initial integrated plan and submitted only at their own gates, while its physical commands remain in force throughout delivery. Publication, fees, actual state carryover and settlement follow the same chronological rules. The event trace identifies each retained submission separately from solver attempts. The initial plan is a perfect-PV planning solution at its modeled boundary, but the execution controller can alter it as actual SOC and retained curtailment recovery evolve. Its realized profit is therefore an information benchmark, not a guaranteed upper bound. Repeated solves can select different schedules when several optima exist, and finite MIP tolerance can permit small objective differences.

Monthly operating profit is additive over delivery products:

$$J_{\mathcal{D}}^{\text{real}} = \sum_{d \in \mathcal{D}} \left[\sum_{a,t} \pi_{d,a,t}^{\text{real}} q_{d,a,t}^{\text{sub}} + \sum_{t \in \mathcal{T}_d} S_t - D_d^{\text{executed}} - C_d^{\text{import}} - C_d^{\text{trading,real}} \right]. \tag{77}$$

The trade ledger preserves submission/publication clocks; daily profit is attributed to delivery date. Fees and auction revenues reconcile both by day and over the complete ledger. Operator fee volume still uses the submission month: the first day's 31 December orders belong to December, even though their revenue appears in 1 January delivery profit. Actual fee mode therefore requires an applicable December tariff and a timestamped initial operator volume state; January volume starts according to the fee calendar, without a daily reset. Wear is charged only on actual executed throughput/history. Per-day initial expectations and realized-minus-expected differences are reported separately; no objective values across overlapping gate solves are added to obtain profit.

With the configured infeasibility policy, a later failed solve can retain the previous physical plan and its already planned current order in integrated mode; the sequential mode retains dispatch and submits zero new trades. No prior plan means the existing failure behavior applies. Actual feedback still enforces physical limits and settles remaining deviations. Failed attempts and fallback choices remain explicit diagnostics.

Resume checks the effective configuration, explicit policy version and common input fingerprints before restoring an atomic completed-midnight checkpoint. The checkpoint contains actual physical/aging state, outstanding priced and unpriced commitments for both relevant books, retained commands, fee state, publication status and the event cursor. Work after that boundary is replayed without double-booking cash or fees. Business decisions rolled back for resume do not survive as submitted history; every actual Gurobi attempt, including discarded work, survives in a separate attempt log with its rollback marker. Native `Model.Runtime` statistics include these attempts, keep reference solves identifiable, and report totals plus mean/minimum/maximum/population standard deviation by strategy and delivery day. Model-build and Python wall times remain separate. Resolved configurations, common-input provenance, daily/monthly KPIs, state projections and comparisons are saved by file type under `outputs/continuous_optimization/farm_<id>/<run_id>/`. The manifest also records the Git revision, but the resume identity is not a hash of every source file. An incompatible policy-version change rejects an older checkpoint; the current exact-PV envelope version is distinct from the preceding controller version.

14 Implemented trading experiments and evidence

The independent-day historical runner is `experiment-gme-oracle`. It reads local cataloged market data, uses the same realized auction price curves for all strategies, forecasts PV using the configured cutoff-available weather, and executes the schedule against measured AC production. Supplying the future realized auction curves at the first decision is an intentional price-oracle assumption: this isolates physical/participation decisions but is not a causal price-forecast backtest and cannot be interpreted as implementable expected profit.

The implemented comparisons are:

- Integrated planning across the currently open auctions versus `current_auction_only`, where only the auction closing now has trade variables. Both carry the same submitted commitments as fixed positions.
- Independent-day participation profiles, comparing all nonempty subsets of DA, IDA1, IDA2 and IDA3 on matched dates, plus a sequential all-auction baseline. Cancellations create neither a solve nor trade variables.
- A frozen-plan replay that measures the value of later forecast updates and replanning while holding previously submitted trades fixed.
- A perfect-PV full-information schedule, and an optional PV-only schedule that removes the battery while preserving the market and farm assumptions. These are operating counterfactuals on the same realized prices and PV data; terminal inventory is reported separately because no terminal energy salvage value is included.
- Continuous daily planning across consecutive delivery days. Each strategy carries actual SOC, degradation history, submitted commitments, transaction-fee volume and cash ledger through local midnight. The daily MILP remains one delivery day at a time; it is not a month-long joint optimization. The January 2026 and August 2026 four-farm configurations compare updating deterministic and 51-member exact-delivery PV forecasts with a daily DA-only perfect-PV reference. August is restricted to 2–31 August because the required 31 July ensemble cycle for 1 August is absent. The opening SOC on 2 August is a new configured initial assumption, not a state carried from 1 August.

The January 2025 participation study uses independent delivery days. The continuous configurations expose inclusive start/end dates and initial SOC; January 2025 starts with DA submission on 31 December, supplied initial SOC 90%, and configured final planned SOC 50%. The January 2026 comparison configs also allow a 40% planned terminal floor. Those are study assumptions, not measured initial conditions or a guarantee on realized terminal energy. The perfect-PV continuous reference solves once at DA (or the first available auction) by default, then retains the plan; a shared flag can request reoptimization at later gates. It is an information benchmark, not a proven upper bound on realized feedback profit: actual controller actions, fees, settlement and final inventory can differ from the planned objective.

Every solve and interval records forecast vintage, position ledger, planned and executed battery flows, SOC, imbalance, throughput/wear, fees and timing. Outputs are grouped by run and file type under `outputs/`. Daily profits are reconciled from submitted trades, actual settlement, executed degradation and fees. The objective values of overlapping auction solves are never summed to represent profit.

15 Price Forecasting Status

The active GME and continuous historical experiments use realized auction clearing curves as an oracle input. They do not use predicted prices or model price uncertainty. The prepared-input optimization route accepts externally supplied point-price forecasts with explicit source, fit and availability timestamps; it rejects prices labeled as realized or oracle values. The generic scenario-tree code also supports synthetic conditional price scenarios, but neither capability establishes a validated historical price-forecasting model. Former retrospective regression, fundamentals and Gaussian-calibration work is archived outside the active pipeline in `/home/rghetti/price_prediction_archive`. Future causal price scenarios need their own evaluation and information-timing protocol.

16 Optional optimized-imbalance formulation

This section describes supported code that is disabled by the shared default. When `optimize_imbalances=true`, replace (21) by

$$\delta^+_{s,t} - \delta^-_{s,t} = g_{s,t}\Delta_t - Q_{s,t}, \tag{78}$$

$$0 \le \delta^+_{s,t} \le M_t y_{s,t}, \tag{79}$$

$$0 \le \delta^-_{s,t} \le M_t(1 - y_{s,t}), \qquad y_{s,t} \in \{0,1\}. \tag{80}$$

Here $M_t = (P^{\text{imp}}+P^{\text{exp}})\Delta_t$, optionally reduced by a configured imbalance-energy cap. The common M_t is valid because physical and contractual net energy use the same directional POI range. The sign binary prevents simultaneous positive/negative deviations, including with negative prices. Add to each scenario's objective

$$\sum_t \left[\lambda^+_{s,t}\delta^+_{s,t} - \lambda^-_{s,t}\delta^-_{s,t} - \kappa(\delta^+_{s,t} + \delta^-_{s,t})\right]. \tag{81}$$

If this mode has zero settlement price, zero penalty and no useful delivery restriction, an optimized contract need not be backed by delivered energy. The January farm-1 scenario study includes configurations with zero assumed planning settlement prices and illustrative absolute-imbalance penalties of 100 and 500 EUR/MWh. Their expected objective averages these terms over members; actual execution uses measured PV and settles deviations at recorded prices. The illustrative planning penalty is not a real market charge. The optimized-imbalance formulation retains all scenario leaves rather than using the exact-delivery envelope reduction; it is a mathematical study mode, not an assertion that deliberate imbalance strategies are permitted.

17 Data provenance, outputs and reproducibility

Farm inputs are local read-only Zarr/Parquet stores. API and workbook records use `data/api/market.duckdb`. Raw/normalized snapshots, source identities, checksums, revision and retrieval metadata support repeated offline studies. Historical publication remains a separate field/convention. Provider acquisition is an explicit user-run command; neither the MILP builder nor the historical experiment downloads missing inputs. DuckDB writers and analysis processes must be coordinated; complete cached days can be studied before an entire acquisition window is complete.

YAML controls study date, calendar validity, source mappings, PV and data policies, BESS and POI characteristics, state, degradation, solver and outputs. One include is resolved relative to the containing YAML, with recursive mapping overrides. Every shipped entry point imports `configs/standard.yaml` directly; experiment YAMLs define workflow-specific settings and overrides. The

standard contains shared operational settings. Explicit null values and replacement lists survive the merge. The `resolve-config` command prints the effective YAML without running a study or accessing provider data.

A single root `farm_id`, optionally overridden with `-farm-id`, selects forecast and physical plant identifiers through typed references resolved after merging. Each farm uses its own weather and measured-power series and Parquet metadata for PV geometry and tracking. Automatic directional POI ratings use declared AC nameplate capacity divided by 1000 as an illustrative MW proxy; per-farm ratings or explicit plant limits override that rule. Battery size and SOC percentages remain common, while automatic battery dimensions resolve from `farms.<id>.battery` for the selected physical plant. Numeric experiment overrides take precedence over these automatic dimensions. The reusable `configs/full_experiments/gme_day.yaml` exposes this workflow.

The current 50% study is configured by `configs/full_experiments/gme_oracle_2025-01-01_farm910_perfect_pv_comparison_terminal_soc_50.yaml`. Its inherited assumptions include a 5 MWh battery, 2.5 MW symmetric power, a 7–93% SOC box, directional combined efficiencies of $\sqrt{0.90} = 0.9486832981$ and a 6.638 MW directional POI. Their product gives 90% AC-to-AC round-trip efficiency for all farms. These are reproducible study settings, not confirmed anonymous-farm hardware.

The shared per-farm hypothetical BESS dimensions are:

Farm	PV AC nameplate (MW)	Battery (MWh)	Symmetric power (MW)
1	0.95328	0.75	0.5
345	1.5246	1.0	0.5
901	4.1600	2.0	1.0
910	6.6380	5.0	2.5

The shared 7–93% SOC limits are converted to energy after selecting capacity; replacement cost scales with that capacity. Throughput economics remain the configured historical USD 115/kWh proxy, 1.0389 USD/EUR conversion and 2,500 reference cycles at 80% depth. They have not been updated to a 2026 equipment quote. With the current efficiencies the derived AC-throughput wear cost is 27.635145 EUR/MWh; no dynamic capacity fade is simulated.

Fresh experiment runs default to `experiment_outputs.overwrite_existing:true`: repeating the same ID replaces artifacts within that run's scope under its configured output root. `resume: true` takes precedence and preserves checkpoints, requiring matching configuration and input fingerprints. Continuous output replacement holds the run lock and retains its inode; unrelated run IDs are preserved.

Each historical run is grouped under `outputs/full_experiments/gme_oracle/<run>/farm_<id>/<day>/` with `plots/`, `csv/`, `json/`, and optional `models/`. The effective configuration, input provenance and artifact manifest accompany auction plans, immutable ledger, actual dispatch, used PV vintages, settlement and objective comparisons. The main KPI distinctions are:

- `total_gain_eur`: actual feedback profit after available settlement;
- `perfect_information_objective_eur`: achieved full-information PV+BESS objective;
- `pv_only_perfect_information_objective_eur`: achieved optional PV-only objective;
- `battery_incremental_operating_value_eur`: the difference in (66);
- initial expectation, realized-minus-expected revenue/profit, and separate executed versus counterfactual degradation components.

Figures preserve delivery boundaries, including next-day local midnight. PV power curves visually

connect successive interval samples; other interval powers and prices are drawn as held values, and SOC uses the actual boundary states. These presentation choices do not change the interval data or energy accounting. Snapshot settlement diagnostics include their different delivery windows and common-horizon comparisons.

Solver settings use relative MIP gap 10^{-4} (0.01%) by default, with no absolute MIP gap configured. Configurable `feasibility_tolerance` and `integrality_tolerance` map to Gurobi's numerical feasibility and integer tolerances. The historical YAML sets both to 10^{-8} to avoid tiny binary-mode leakage while preserving the independent validator's physical checks. The shared time limit is 120 seconds per solve with 10 solver threads. LP model files are not generated by default; a study may explicitly enable model export, and infeasibility diagnostics have their own switch. Objective comparisons use achieved `ObjVal`, not a claim of exact global optimality beyond the recorded solver tolerance. Status, relative gap, model sizes and timing remain recorded. Before a schedule is used, independent validation recomputes physical balances, limits, native product consistency, information sharing, degradation and objective components. For optimized imbalance, the modeled cash flow, penalty and CVaR are independently recomputed from the solver's positive and negative imbalance variables. Those variables are also checked against the physical grid-minus-position balance at every interval. The validation report records the maximum EUR difference between split-variable and physically reconstructed accounting, without relaxing the energy-balance tolerance when a large penalty scales small MWh residuals. For continuous replay, the reported solver objective is checked against the independently recomputed objective using the absolute validation tolerance plus 10^{-8} times the larger objective magnitude; this relative term is distinct from the MIP gap and applies to that scalar comparison. It can be set via `execution.validation_objective_relative_tolerance`. Continuous objective components and, when available, the objective expression evaluated at the extracted solution use an absolute EUR allowance plus a separate relative allowance, checked for each scenario. The report records the applied tolerances and objective residuals. Ledger and physical reporting use the same configured absolute and relative numerical comparison convention. Nonfinite values, missing inputs and temporal/identity contracts still require exact validity. The solver's reported objective is not replaced by the recomputed value. In continuous replay, if the extracted objective expression and all independent physical and component checks pass, a discrepancy in the solver's reported objective is saved and displayed as a warning rather than stopping the replay. A missing or mismatched extracted expression does not receive that exception. When CVaR is enabled, the returned threshold and excess variables are checked against the scenario-loss epigraph and used to reconstruct the objective, including for a time-limited incumbent. Execution separately checks actual SOC and power/POI protection. Software consistency does not resolve uncertain farm metadata, forecast bias or historical publication vintages.

18 Current limitations and extension boundaries

The current empirical workflows use daily perfect-price planning with either one deterministic weather trajectory or an optional ensemble, in independent days or consecutive continuous operation, with approximate geometry and an idealized interval-average controller. Ensemble weather members are coherent input paths, but they are not calibrated probabilistic guarantees; exact scenario feasibility does not guarantee feasible delivery under unseen PV. Measured AC is effective observed production, not an independently estimated unconstrained irradiance-to-generation potential; hidden plant curtailment/faults in source production are not identified automatically. Configured cleaning and fallback choices remain part of the experiment.

The model currently omits continuous trading, order acceptance, market impact, multiple-farm network coupling, ancillary services, intrainterval dynamics, ramping/startup constraints, dynamic capacity loss, calendar aging, thermal battery dynamics, fixed operating costs and investment

cash-flow optimization. The generic `backtest` CLI remains synthetic-only; the separate historical `experiment-gme-oracle`, `experiment-participation` and `experiment-continuous` workflows are the implemented manual-data replays. Future uncertain-price evaluations need their own causal inputs and comparison protocol rather than relabeling the current oracle results.

The original implementation-status report is retired. This source and its PDF are the maintained mathematical description; run manifests and saved solver results are the numerical evidence for particular configurations. Practical commands and configuration examples are in the root `README.md`.

References

[1] B. Xu, J. Zhao, T. Zheng, E. Litvinov and D. S. Kirschen, *Factoring the Cycle Aging Cost of Batteries Participating in Electricity Markets*. Author preprint and published paper links retained by the repository: https://arxiv.org/abs/1707.04567 and https://doi.org/10.1109/TPWRS.2017.2733339.

[2] pvlib Python documentation. The installed library implements the solar-position, single-axis tracking, isotropic transposition, Faiman and PVWatts functions called by `forecasting/models/pv.py`. https://pvlib-python.readthedocs.io/.

[3] Gestore dei Mercati Energetici, Technical Rule 03 rev. 13 MPE, valid from 29 September 2025, and rev. 14 MPE, valid from 13 January 2026. The project uses their gate/result timing with explicit local study settings; DA publication at 13:30 is identified as an assumption, while recorded times are preferred. GME's 29 September 2025 notice describes 15-minute MGP products from delivery date 1 October 2025. https://www.mercatoelettrico.org/Portals/0/Documents/en-US/20250930DTF03rev13MPE_En.pdf https://gme.mercatoelettrico.org/Portals/0/Documents/en-US/20260113DTF03rev14MPE_en.pdf https://www.mercatoelettrico.org/en-us/Home/NoticesandAnnouncements/NoticesandAnnouncementsME?id=7101

[4] Repository source files listed in Section 1 and the effective configuration saved with each run are authoritative for implemented behavior. Conceptual references do not replace inspection of those inputs and policies.